



**National University of Sciences and Technology (NUST)**  
**School of Electrical Engineering and Computer Science**

## **CS: Data Structures and Algorithms**

### **Lab 01: Introduction to Version Control (Git) and Simple Test**

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## Lab 1: Arrays, Basic OOP and Git

### Lab Tasks

#### Task 1:

Create an array named numbers containing 2, 4, 6, 8 and 10. Change the third element to 7. Use a for loop to display all five values, separated by spaces. Predict the output before running your program.

#### Solution:

The output would be 2 4 7 8 10

```
raojawad@Raos-MacBook-Pro ~/D
$ ./lab1_task1
2 4 7 8 10 %
```

#### Task 2:

Declare `int numbers[5];` and initialise `int total = 0;`. Use one loop to read five integers into the array and a second loop to add them to total. Display the total. Assume the user enters exactly five valid integers whose sum fits in `int`.

| Test input  | Expected total | Actual total |
|-------------|----------------|--------------|
| 1 2 3 4 5   | 15             | 15           |
| 0 0 0 0 0   | 0              | 0            |
| -2 4 -1 0 3 | 4              | 4            |

For the first test, record `i`, `numbers[i]` and total after each addition. If a result differs from your prediction, check the loop bounds and the initial value of total.

#### Solution:

For the first test.. the `i` went 0 then 1 then 2 then 3 and then 4 and the total went 1 then 3 then 6 then 10 and finally 15.



```
raojawad@Raos-MacBook-Pro ~/Documents/Developer/3rd semester/DSA/DSA-lab1 <main>
$ ./lab1_task2
Enter number 1: 1
Enter number 2: 2
Enter number 3: 3
Enter number 4: 4
Enter number 5: 5
the value of i 0
the value of total 1
the value of i 1
the value of total 3
the value of i 2
the value of total 6
the value of i 3
the value of total 10
the value of i 4
the value of total 15
Total: 15
```

```
raojawad@Raos-MacBook-Pro ~/Documents/Developer/3rd semester/DSA/DSA-lab1 <main>
$ ./lab1_task2
Enter number 1: 0
Enter number 2: 0
Enter number 3: 0
Enter number 4: 0
Enter number 5: 0
Total: 0
```

```
raojawad@Raos-MacBook-Pro ~/Documents/Developer/3rd semester/DSA/DSA-lab1 <main>
$ ./lab1_task2
Enter number 1: -2
Enter number 2: 4
Enter number 3: -1
Enter number 4: 0
Enter number 5: 3
Total: 4
```

### Task 3:

Define a class named Student with two public integer data members, rollNumber and marks. Add a public member function void display() that prints both values with labels on separate lines. Define the class above main().

Inside main(), create objects s1 and s2. Assign and display the following values using the dot operator and display():

| Object | rollNumber | marks |
|--------|------------|-------|
| s1     | 1          | 75    |



s2

2

90

Write the expected output before running. Then change only s1.marks to 80 and display both objects again. Predict whether s2.marks changes, and explain the result. Keep this experiment in your final program.

### Solution:

S2.marks WILL change because s2.marks does not share memory with s1.marks.

```
raojawad@Raos-MacBook-Pro ~/Documents/Developer/3r
$ ./lab1_task3
Roll Number: 1
Marks: 75
Roll Number: 2
Marks: 90
Roll Number: 1
Marks: 80
Roll Number: 2
Marks: 90
```

### Task 4:

Read 8 integers into an array. Find and display the largest and smallest values and their indices. If a value occurs more than once, report its first occurrence.

### Solution:

```
raojawad@Raos-MacBook-Pro ~/Documents/Developer/3r
$ ./lab1_task4
Enter number 1: 2
Enter number 2: 4
Enter number 3: 7
Enter number 4: 3
Enter number 5: 5
Enter number 6: 2
Enter number 7: 4
Enter number 8: 2
The largest number is: 7
Index of the largest number is: 2
The smallest number is: 2
Index of the smallest number is: 2
```



### Task 6:

Read 6 integers into an array. Reverse the elements **without using another array**, then display the updated array.

Example: {3, 7, 2, 9, 4, 1} becomes {1, 4, 9, 2, 7, 3}.

### Solution:

```
Index of the smallest number is: 2
raojawad@Raos-MacBook-Pro ~/Documents
$ ./lab1_task6
Enter number 1: 1
Enter number 2: 2
Enter number 3: 3
Enter number 4: 4
Enter number 5: 5
Enter number 6: 6
Reversed array: 6 5 4 3 2 1 %
```

### Task 7:

Read 10 integers into an array. Move the first occurrence of each distinct value into the beginning of the same array, preserving the original order. Display these unique values and their count. Do not use another array.

Example: {4, 2, 4, 7, 2, 9, 7, 1, 9, 1} produces {4, 2, 7, 9, 1}, with a count of 5.

The array's physical capacity remains 10; only the first 5 positions represent the result.

### Solution:



```
rajand@rajand:~/Documents$ ./lab1_task7
Enter number 1: 1
Enter number 2: 6
Enter number 3: 4
Enter number 4: 8
Enter number 5: 3
Enter number 6: 2
Enter number 7: 1
Enter number 8: 4
Enter number 9: 6
Enter number 10: 7
Unique values: 1 6 4 8 3 2 7
Count of unique values: 7
```

### Task 8:

Create a folder named dsa-lab01 and place your tasks inside it. Open a terminal in that folder. Git setup is to be demonstrated by the lab engineer.

- git init
- git status
- git add task2.cpp task3.cpp task4.cpp
- git commit -m "Add array and Student class exercises"
- Create README.md with your name, registration number, one sentence about each program, and the three Task 3 test results. Then run:
  - git add README.md
  - git commit -m "Document programs and test results"
  - git status
  - git log --oneline
  - git add stages selected files. git commit records the staged version locally. git status shows file changes; git log shows commits. A commit does not upload files to GitHub. Stage only the named source and README files, not executables.
- If Git asks for an identity, the instructor will help you run git config user.name "Your Name" and git config user.email "your-email@example.com" with your own details. These settings apply to this repository.

### Solution:



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```
raojawad@Raos-MacBook-Pro ~/Documents/Developer/3rd semester/DSA/DSA-lab1 <main>
$ git log --oneline
raojawad@Raos-MacBook-Pro ~/Documents/Developer/3rd semester/DSA/DSA-lab1 <main>
$ git add README.md
raojawad@Raos-MacBook-Pro ~/Documents/Developer/3rd semester/DSA/DSA-lab1 <main>
$ git commit -m "edited the readme a bit"
[main 1f6d183] edited the readme a bit
1 file changed, 8 insertions(+), 1 deletion(-)

raojawad@Raos-MacBook-Pro ~/Documents/Developer/3rd semester/DSA/DSA-lab1 <main>
$ git status
On branch main
Changes not staged for commit:
  (use "git add <file>..." to update what will be committed)
  (use "git restore <file>..." to discard changes in working directory)
        modified:   lab1_task2.cpp

Untracked files:
  (use "git add <file>..." to include in what will be committed)
        .vscode/
        lab1_task1
        lab1_task1.cpp
        lab1_task1.dSYM/
        lab1_task2
        lab1_task2.dSYM/
        lab1_task3
        lab1_task3.dSYM/
        lab1_task4
        lab1_task4.dSYM/
        lab1_task6
        lab1_task6.cpp
        lab1_task6.dSYM/
        lab1_task7
        lab1_task7.cpp
        lab1_task7.dSYM/
        practice
        practice.cpp
        practice.dSYM/

no changes added to commit (use "git add" and/or "git commit -a")

1f6d183 (HEAD -> main) edited the readme a bit
fbc8201 Document programs and test results
0af4a34 Add array and Student class exercises
(END)
```

### Deliverables



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- Submit the .cpp files, README.md and a short PDF report on LMS (or the repository link if permitted). The report should contain your name and registration number.